



SANT'ANNA WORKSHOP ON CAUSAL INFERENCE

Covariates in DiD

OLS specs, IPW, OR, DR

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Recall: the 2×2 is DiD — six equivalent forms

$$1. \underbrace{\left[\mathbb{E}[Y|D=1, t] - \mathbb{E}[Y|D=1, t-1] \right]}_{\text{first difference}} - \underbrace{\left[\mathbb{E}[Y|D=0, t] - \mathbb{E}[Y|D=0, t-1] \right]}_{\text{first difference}}$$

$$2. \underbrace{\left[\mathbb{E}[Y|D=1, t] - \mathbb{E}[Y|D=0, t] \right]}_{\text{group difference}} - \underbrace{\left[\mathbb{E}[Y|D=1, t-1] - \mathbb{E}[Y|D=0, t-1] \right]}_{\text{group difference}}$$

$$3. Y = \alpha_1 + \alpha_2 \text{Post} + \alpha_3 D + \delta (\text{Post} \times D) + \varepsilon \quad \text{saturated regression}$$

$$4. Y = \alpha_1 + \tau_t + \sigma_i + \delta (\text{Post} \times D) + \varepsilon \quad \text{two-way fixed effects}$$

$$5. \Delta Y_i = \alpha_1 + \delta D_i + \varepsilon_i \quad \text{first-difference / horizontal regression}$$

$$6. \Delta \bar{Y}_G = \alpha_1 + \delta \text{Post}_G + \varepsilon_G \quad \text{group / vertical regression}$$

All six are **numerically identical**.

The 2×2 DiD estimand

$$\underbrace{\hat{\delta}_{\text{DiD}}}_{\text{what we estimate}} = \underbrace{\text{ATT}}_{\text{what we want}} + \underbrace{\left[\mathbb{E}[\Delta Y(0) \mid D=1] - \mathbb{E}[\Delta Y(0) \mid D=0] \right]}_{\text{PT bias}}$$

Treatment = college education. Outcome = wages.
Parallel trends requires the PT bias term to equal zero.

A thought experiment

Group	Untreated wage trend $\Delta Y(0)$
Male workers	+10 per year
Female workers	+8 per year

Case 1: $D=1$ is 75% male. $D=0$ is 75% male.

Will parallel trends hold?

Case 1: show the work

	Calculation	Result
$\mathbb{E}[\Delta Y(0) \mid D=1]$	$0.75 \times 10 + 0.25 \times 8$	9.5
$\mathbb{E}[\Delta Y(0) \mid D=0]$	$0.75 \times 10 + 0.25 \times 8$	9.5

Parallel trends holds? **Yes.**

$$\hat{\delta}_{\text{DiD}} = \text{ATT} \quad \checkmark$$

Case 2: unequal composition

	% male
$D=1$ (college educated)	75%
$D=0$ (no college)	25%

Same wage trends as before (+10 male, +8 female).

Will parallel trends hold now?

Case 2: show the work

	Calculation	Result
$\mathbb{E}[\Delta Y(0) \mid D=1]$	$0.75 \times 10 + 0.25 \times 8$	9.5
$\mathbb{E}[\Delta Y(0) \mid D=0]$	$0.25 \times 10 + 0.75 \times 8$	8.5

Parallel trends holds? **No.**

$\hat{\delta}_{\text{DiD}} \neq \text{ATT}$ because PT bias = $9.5 - 8.5 = 1.0 \neq 0$

Why? Imbalance on a trend-causing covariate

DiD does **not** require the two groups to look the same on *levels*.

DiD **does** require the same $Y(0)$ *trends*.

Imbalance in a covariate that *causes* $Y(0)$ trends
will **mechanically** break parallel trends.

Fix: control for sex

If sex is the only covariate driving differential trends,
then *conditioning on sex* removes the PT bias.

Several ways to do this:

1. TWFE saturated regressions
2. Inverse probability weighting (IPW)
3. Outcome regression (OR)
4. Doubly robust (DR)

We will discuss all four.

For conditioning to work: conditional PT

We need parallel trends *within* each group:

1. Male treated workers have the *same* $Y(0)$ trend as male control workers
2. Female treated workers have the *same* $Y(0)$ trend as female control workers

Conditional PTA: $\mathbb{E}[\Delta Y(0) \mid D=1, X] = \mathbb{E}[\Delta Y(0) \mid D=0, X]$ for all X

The standard TWFE with additive X

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \varepsilon_{it}$$

Two hidden assumptions:

1. Returns to X are **constant over time**
2. Treatment effect is **the same for all X**

Fatal problem: if X_i is time-invariant, it is collinear with α_i and drops out — whether you use unit fixed effects *or* a saturated dummy specification. An additive, separable X cannot deliver conditional parallel trends if the covariate is absorbed before it can do any work.

What if returns to X change over time?

Example: discrimination against women is not constant — $\beta_{\text{post}} \neq \beta_{\text{pre}}$.

Allow time-varying returns in the DGP:

$$Y_{it}(0) = \alpha_i + \gamma_t + \beta_t X_i$$

Write out the **post** and **pre** equations for unit i :

$$Y_{i,\text{post}}(0) = \alpha_i + \gamma_{\text{post}} + \beta_{\text{post}} X_i$$

$$Y_{i,\text{pre}}(0) = \alpha_i + \gamma_{\text{pre}} + \beta_{\text{pre}} X_i$$

Subtract (post – pre):

$$\Delta Y_i(0) = \underbrace{(\gamma_{\text{post}} - \gamma_{\text{pre}})}_{\Delta\gamma} + \underbrace{(\beta_{\text{post}} - \beta_{\text{pre}})}_{\Delta\beta} \cdot X_i$$

Now take expectations by group

Starting from $\Delta Y_i(0) = \Delta\gamma + \Delta\beta \cdot X_i$, condition on treatment status:

$$E[\Delta Y(0) \mid D=1] = \Delta\gamma + \Delta\beta \cdot E[X \mid D=1]$$

$$E[\Delta Y(0) \mid D=0] = \Delta\gamma + \Delta\beta \cdot E[X \mid D=0]$$

Parallel trends bias:

$$\underbrace{E[\Delta Y(0) \mid D=1] - E[\Delta Y(0) \mid D=0]}_{\text{PT bias}} = \Delta\beta \cdot (E[X \mid D=1] - E[X \mid D=0])$$

PT fails whenever **both**: returns shift ($\Delta\beta \neq 0$) **and** groups are imbalanced on X .

Fix for Bias #1: interact $\text{Post} \times X$

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \gamma (\text{Post}_t \times X_i) + \varepsilon_{it}$$

γ absorbs any shift in returns to X between pre and post (Spec B in the lab).

```
reg re i.post##i.ever_treated ///  
    i.post##c.age i.post##c.agesq i.post##c.agecube ///  
    i.post##c.educ i.post##c.educsq ///  
    i.post##i.marr i.post##i.nodegree i.post##i.black i.post##i.hisp ///  
    i.post##c.re74 i.post##i.u74, robust
```

neous Treatment Effects

What if the effect of college differs by sex?

Previously we worried about **trends in** $Y(0)$ being unequal across groups.

Now the concern is different:

$$\mathbb{E}[Y(1) - Y(0) \mid \text{Male}] \neq \mathbb{E}[Y(1) - Y(0) \mid \text{Female}]$$

Do the rich and the poor get the same return to college?

Do workers in cities and towns get the same return?

If not, **and** the treatment group is imbalanced on these covariates,
the 2×2 DiD will not recover the ATT.

Proof, step 1 — the TWFE model and conditional PT

The TWFE model with one covariate X :

$$Y_{it} = \alpha_i + \gamma_t + \delta (T_t \times D_i) + \beta X_i + \varepsilon_{it}$$

Conditional parallel trends says, for each value of X :

$$\mathbb{E}[Y(0)_{\text{post}} - Y(0)_{\text{pre}} \mid D=1, X] = \mathbb{E}[Y(0)_{\text{post}} - Y(0)_{\text{pre}} \mid D=0, X]$$

Rearranged — the missing counterfactual for the treated is identified:

$$\mathbb{E}[Y(0)_{\text{post}} \mid D=1, X] = \mathbb{E}[Y_{\text{pre}} \mid D=1, X] + \mathbb{E}[Y_{\text{post}} \mid D=0, X] - \mathbb{E}[Y_{\text{pre}} \mid D=0, X]$$

Proof, step 2 — plug TWFE in: the slopes look the same

From the TWFE model, observed conditional means:

$$\mathbb{E}[Y_{\text{pre}} \mid D=1, X] = \alpha_1 + \alpha_3 + \beta X$$

$$\mathbb{E}[Y_{\text{post}} \mid D=0, X] = \alpha_1 + \alpha_2 + \beta X$$

$$\mathbb{E}[Y_{\text{pre}} \mid D=0, X] = \alpha_1 + \beta X$$

Counterfactual for treated (step 1 identity):

$$\mathbb{E}[Y(0)_{\text{post}} \mid D=1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \beta X$$

Treated outcome, post: $\mathbb{E}[Y(1)_{\text{post}} \mid D=1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \delta + \beta X$

Both potential outcomes carry the *same* β . Subtract: $ATT = \delta$. *Suspiciously clean.*

Proof, step 3 — what if $Y(1)$ and $Y(0)$ have different slopes in X ?

Let the slopes differ: write β_1 for the $Y(1)$ slope, β_2 for $Y(0)$:

$$\mathbb{E}[Y(1)_{\text{post}} \mid D=1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \delta + \beta_1 X$$

$$\mathbb{E}[Y(0)_{\text{post}} \mid D=1, X] = \alpha_1 + \alpha_2 + \alpha_3 + \beta_2 X$$

True ATT given X :

$$\text{ATT}(X) = \delta + (\beta_1 - \beta_2) X$$

- If $\beta_1 \neq \beta_2$, the treatment effect *depends on* X
- TWFE fits one β , not two — so $\hat{\delta}$ recovers ATT only if $\beta_1 = \beta_2$

Proof, step 4 — TWFE imposes homogeneous δ in X

Assumption: TWFE-with- X requires the treatment effect to be the same at every value of X .

- If X is **sex**: effect must be equal for men and women
- If X is **income**: effect must be equal at \$1 and at \$1,000,000
- If X is **age**: effect must be equal at 25 and at 65

OLS doesn't warn you that you're making this assumption. The math does.
If effects actually vary with X , TWFE averages across the variation
and returns a weighted parameter you didn't ask for.

Fix for Bias #2: saturate treatment with X

1. **Run a saturated regression on first differences** — interact treatment with every X
2. **Predict $\Delta Y(1)$ and $\Delta Y(0)$ for each unit** — average the gap in the treated group

```
* 1. Saturated regression on first differences
reg dy i.ever_treated ///
    i.ever_treated#c.age i.ever_treated#c.agesq ///
    i.ever_treated#c.educ i.ever_treated#c.educsq ///
    i.ever_treated#i.marr i.ever_treated#i.nodegree ///
    i.ever_treated#i.black i.ever_treated#i.hisp ///
    i.ever_treated#c.re74 i.ever_treated#i.u74 ///
    $X, robust

* 2. Predict potential outcomes; average the gap in the treated group
replace ever_treated = 1
quietly predict dy_hat_1, xb
replace ever_treated = 0
quietly predict dy_hat_0, xb
replace ever_treated = evt_orig // restore
gen tau_sat = dy_hat_1 - dy_hat_0
quietly sum tau_sat if ever_treated == 1 // ATT = r(mean)
```

Bias #3: time-invariant covariates get absorbed

If you need X_i for conditional PT, but X_i doesn't change over time,
TWFE will **delete it**.

TWFE “demeans” the data. The mean of a constant is itself.
So $X_i - \bar{X}_i \equiv 0$ — it drops out of the regression entirely.

Fix: include $X_i \times \text{Post}_t$ instead. The Post interaction is time-varying,
so it survives demeaning and can do the work of conditioning on X .

Three biases — one family of fixes

Bias	What TWFE imposes	Fix
#1 Trend imbalance	β constant pre and post	$\text{Post} \times X$
#2 Heterogeneous TE	δ same at every X	$D \times X$ (saturate treatment)
#3 Time-invariant X	X_i survives demeaning	$\text{Post} \times X$ (same fix as #1)

Most researchers include additive X and call it a day. That is the wrong spec. To fix all three you need both $\text{Post} \times X$ and $D \times X$. It is rare to see this in published work.

IPW: reweight the controls to look like the treated

$$\widehat{\text{ATT}}_{\text{IPW}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)} \Delta Y_i$$

$$\hat{p}(X_i) = \widehat{\text{Pr}}(D_i = 1 \mid X_i) \quad (\text{propensity score}) \quad \Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$$

IPW: five steps

1. **Estimate $\hat{p}(X)$.** Logit of D_i on X_i .
2. **Check overlap.** Plot $\hat{p}$ by treatment status. Trim if needed.
3. **Construct weights.** $w_i = (D_i - \hat{p}(X_i)) / (1 - \hat{p}(X_i))$.
4. **Weighted average.** $\widehat{ATT} = \sum w_i \Delta Y_i / n_T$.
5. **Standard error.** Bootstrap or influence function.

Overlap is non-negotiable

If $\hat{p}(X_i) \rightarrow 1$ for a control unit, its weight $\rightarrow \infty$.

No overlap = no valid counterfactual.

This is a data problem. No estimator fixes it.

OR: model the trend, impute the counterfactual

$$\widehat{\text{ATT}}_{\text{OR}} = \frac{1}{n_T} \sum_{D_i=1} \left[\Delta Y_i - \hat{\mu}_0(X_i) \right]$$

$$\hat{\mu}_0(X_i) = \hat{E}[\Delta Y \mid D = 0, X] \quad \text{estimated on controls only}$$

OR: five steps

1. **First-difference.** $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$.
2. **Fit outcome model.** Regress ΔY on X *using controls only*.
3. **Predict counterfactual.** Evaluate $\hat{\mu}_0(X_i)$ for each treated unit.
4. **Mean gap.** $\text{ATT} = \overline{\Delta Y}^T - \overline{\hat{\mu}_0(X)}^T$.
5. **Standard error.** Bootstrap.

DR: two chances to be right

$$\widehat{\text{ATT}}_{\text{DR}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \underbrace{\frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)}}_{\text{IPW correction}} \cdot \underbrace{\left[\Delta Y_i - \hat{\mu}_0(X_i) \right]}_{\text{OR residual}}$$

DR is consistent if *either* model is right — not both

$\hat{\mu}_0$ correct	$\hat{p}$ correct	Consistent?
✓	✓	✓
✓	×	✓
×	✓	✓
×	×	No

“Doubly robust” $\neq$ doubly unbeatable.

Lab: ten estimators, one benchmark

Labs/Lalonde-Covariates/lalonde_all_specs.do

NSW treated + CPS controls · Pre = 1975, Post = 1978

RCT benchmark: **\$1,794**

Spec 0 (Naive) · Spec A (Additive X) · Spec B ($\text{Post} \times X$) · Spec C (Saturated)

OR by hand · OR drdid · IPW by hand · IPW drdid · DR by hand · DR drdid

A Monte Carlo to rank the estimators

Estimators: TWFE, OR, IPW, DR

DGPs: 4 (vary whether OR model and pscore are correct)

n : 1,000 Replications: 10,000

Report: Bias, RMSE, SE, coverage, CI length

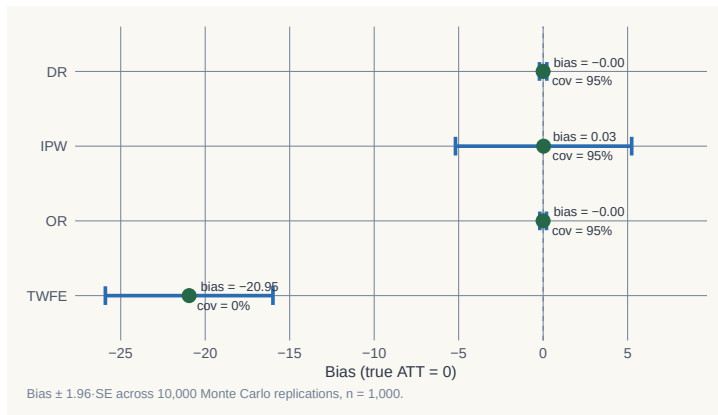
When does DR pay off? When does it break?

DGP 1: both models correct

	Bias	RMSE	SE	Coverage	CI length
TWFE	−20.95	21.12	2.53	0.000	9.91
OR	−0.001	0.10	0.10	0.950	0.40
IPW	+0.026	2.77	2.66	0.952	10.44
DR	−0.001	0.11	0.11	0.947	0.41

TWFE: coverage zero. OR and DR: tight on truth. IPW: unbiased but wide.

DGP 1: the sampling distribution

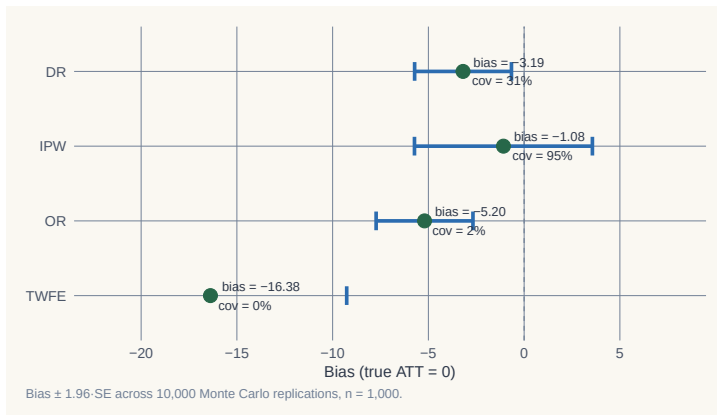


DGP 4: both models wrong

	Bias	RMSE	SE	Coverage	CI length
TWFE	−16.38	16.54	3.63	0.000	14.22
OR	−5.20	5.36	1.29	0.015	5.05
IPW	−1.08	2.66	2.37	0.949	9.31
DR	−3.19	3.45	1.29	0.308	5.07

DR undercovers at 0.31. IPW wins. Doubly robust $\neq$ doubly unbeatable.

DGP 4: the sampling distribution



What the Monte Carlo teaches

- **TWFE + additive X** : coverage zero in every DGP. **Stop using it as a default.**
- **OR**: efficient when right. Coverage 1.5% when wrong.
- **IPW**: less efficient, but robust to outcome misspecification.
- **DR**: best when one model is right. No better than IPW when both are wrong.

Adding X doesn't make your DiD credible

Adding X shifts the credibility question—
from **“does PT hold?”**
to **“is my model of $Y(0)$ or $p(X)$ correct?”**

You still have to answer it.

Now let's see it with real data

LaLonde (1986) · Dehejia & Wahba (2002)

A famous dataset where we **know the true effect**.

The benchmark: **\$1,794**.

LaLonde (1986): the RCT found $\approx \$800$

- Job training program (National Supported Work Demonstration)
- Randomized — treated and control groups assigned by lottery
- RCT estimate: earnings effect $\approx \$800$

A clean causal estimate. The benchmark exists.

LaLonde: dropped the RCT controls, used CPS instead

- Replaced the randomized controls with observational data (CPS, PSID)
- Re-ran every evaluator method from the literature
- Results ranged from **−\$16,000 to +\$3,000**

The true effect was \$800. Not one method recovered it.

“The available evidence calls into serious question the ability of nonexperimental evaluators. . . ”

Empirical labor economics faces an evaluation problem.

Dehejia & Wahba (2002): propensity scores get close

- Need **two pre-treatment years** for all datasets — uses a shorter sample
- RCT benchmark on this shorter sample: **\$1,794**
- Applied propensity score matching on the non-experimental sample
- Result: estimates close to \$1,794

Selective use of the data + right method = nearly recovered the truth.

Our exercise: DiD with and without covariates

NSW treated + CPS controls · Pre = 1975 · Post = 1978

We know the effect: **\$1,794**

Let's see which specs recover it — and which don't.

Naive TWFE: \$3,621 — almost double the truth

```
reg re i.post##i.ever_treated, robust
```

Estimator	ATT	vs benchmark
Naive TWFE	\$3,621	+\$1,827

No covariates — the CPS controls trend very differently from NSW treated.

Additive X : still \$3,621

```
reg re i.post##i.ever_treated $X, robust
```

Estimator	ATT	vs benchmark
Additive X	\$3,621	+\$1,827

Adding X as time-invariant regressors does nothing here. The unit FE absorbs the level differences. The trend bias remains.

Post $\times$ X: \$1,711 — now in the right range

```
reg re i.post##i.ever_treated ///  
    i.post##c.age i.post##c.agesq i.post##c.agecube ///  
    i.post##c.educ i.post##c.educsq ///  
    i.post##i.marr i.post##i.nodegree ///  
    i.post##i.black i.post##i.hisp ///  
    i.post##c.re74 i.post##i.u74, robust
```

Estimator	ATT	vs benchmark
Post $\times$ X	\$1,711	−\$83

Saturated TWFE: \$1,770

```
* First-difference, then regress dy on X + D x X interactions
reg dy $X ///
    i.ever_treated#c.age i.ever_treated#c.agesq ///
    i.ever_treated#c.educ i.ever_treated#c.re74 ///
    i.ever_treated, robust
* Predict counterfactuals; ATT = mean(tau_sat) for treated
```

Estimator	ATT	vs benchmark
Saturated TWFE	\$1,770	−\$24

OR (HIT 1997): same answer as saturated TWFE

```
* Fit trend model on controls only
reg dy $X if ever_treated == 0, robust
predict dy_hat
* ATT = mean gap for treated
gen delta_hat = dy - dy_hat if ever_treated == 1
```

Estimator	ATT	vs benchmark
OR by hand / drdid reg	\$1,770	−\$24

Saturated TWFE and OR are algebraically equivalent in the 2-period case.

IPW (Abadie 2005): \$1,861

```
drdid re $X, time(year) ivar(id) tr(ever_treated) ipw
```

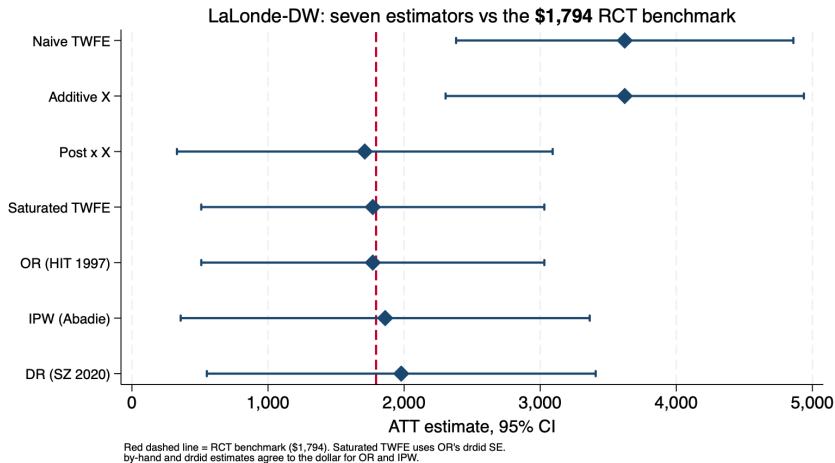
Estimator	ATT	vs benchmark
IPW	\$1,861	+\$67

DR (Sant'Anna–Zhao 2020): \$1,979

```
drdid re $X, time(year) ivar(id) tr(ever_treated) dripw
```

Estimator	ATT	vs benchmark
DR	\$1,979	+\$185

All seven estimators vs the \$1,794 benchmark



Covariates are not robustness checks — they are design

- People say: *“if adding covariates changes results, be suspicious.”*
- That gets it backwards.
- We **know** the effect is \$1,794.
- Without covariates, or with additive covariates, we get \$3,621 — *wrong*.
- The correct specs recover \$1,770–\$1,979 — *right*.

Covariates fix broken parallel trends. Changing your result is not a bug. It is the design working.

The question is always
“did I get the model right?”
